

ECE 3317

Homework #2

Answers

- 1) If $\underline{B}(x, y, z) = xy^2\hat{x} + xy^2z^3\hat{y} + x^2yz^2\hat{z}$, find $\nabla \times \underline{B}$ and $\nabla \cdot \underline{B}$.

$$\nabla \times \underline{B} = (x^2z^2 - 3xy^2z^2)\hat{x} - (2xyz^2)\hat{y} + (y^2z^3 - 2xy)\hat{z}$$

$$\nabla \cdot \underline{B} = y^2 + 2xyz^3 + 2x^2yz$$

- 5) Given that $\underline{E}(x, y, z, t) = 2x\hat{x} + y\hat{y} + 3z\hat{z}$ is a field that exists in space, does $\underline{B}(x, y, z, t)$ vary with time? Justify your answer. Repeat for $\underline{E}(x, y, z, t) = 2y\hat{x} + z\hat{y} + 3x\hat{z}$.

$$\underline{E}(x, y, z, t) = 2x\hat{x} + y\hat{y} + 3z\hat{z}, \quad \nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t} = 0, \quad \underline{B}(x, y, z, t) \text{ does not vary with time.}$$

$$\underline{E}(x, y, z, t) = 2y\hat{x} + z\hat{y} + 3x\hat{z}, \quad \nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t} \neq 0, \quad \underline{B}(x, y, z, t) \text{ varies with time.}$$

- 6) Given that $\underline{D}(x, y, z, t) = 2xy^2\hat{x} + 3x\hat{y} - x^2y\hat{z}$ in some region, what is the charge density in that region?

$$\rho_v = \nabla \cdot \underline{D} = 2y^2$$

- 7) For the following sinusoidal electric fields in free space, find the magnetic field $\underline{H}(x, y, z, t)$.

(a) $\underline{E}(x, y, z, t) = E_0(x\hat{x} + \cos(2\pi z)\hat{y} + 2z\hat{z})\sin \omega t$

(b) $\underline{E}(x, y, z, t) = E_0\hat{z}\sin[2\pi \times 10^6 t + 2\pi(2x + 3y)]$

(c) $\underline{E}(x, y, z, t) = \hat{y}E_0x^2\cos \omega t$.

(a) $\underline{H}(x, y, z, t) = \hat{x}\frac{2\pi E_0}{\mu\omega}\sin(2\pi z)\cos \omega t$

(b) $\underline{H}(x, y, z, t) = (-3\hat{x} + 2\hat{y})\frac{E_0}{10^6\mu}\sin 2\pi(10^6 t + 2x + 3y)$

(a) $\underline{H}(x, y, z, t) = -\hat{z}\frac{2E_0}{\mu\omega}\sin \omega t$

8) If the phasor-domain fields in a region of space are

$$\underline{\underline{E}} = (\hat{x} - j\hat{y})e^{-j(2\pi)z}$$

$$\underline{\underline{H}} = 0.002654(\hat{x} - j\hat{y})e^{-j(2\pi)z},$$

find the complex Poynting vector $\underline{\underline{S}}$ and the time average Poynting vector $\langle \underline{\underline{S}}(t) \rangle$.

$$\underline{\underline{S}} = \frac{1}{2}(\underline{\underline{E}} \times \underline{\underline{H}}^*) = \hat{z}(j0.002654)$$

$$\langle \underline{\underline{S}}(t) \rangle = \text{Re}\{\underline{\underline{S}}\} = \underline{\underline{0}}$$

9) Consider a coaxial cable of inner radius a and outer radius b , filled with a lossless dielectric material. The phasor-domain fields inside of the cable ($a < \rho < b$) for a time-harmonic wave traveling down the cable are given in cylindrical coordinates as

$$\underline{\underline{E}} = \hat{\rho}\left(\frac{1}{\rho}\right)e^{-jkz}$$

$$\underline{\underline{H}} = \hat{\phi}\left(\frac{1}{\rho}\right)\frac{1}{\eta_0\sqrt{\epsilon_r}}e^{-jkz}.$$

Calculate the total time-average power (in Watts) flowing in the z direction down the cable. (Note: The real-valued wavenumber k is the wavenumber of the medium filling the coax, $k = \omega\sqrt{\mu\epsilon}$, but its value should not affect your answer.)

$$P = \frac{\pi}{\eta_0\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$